

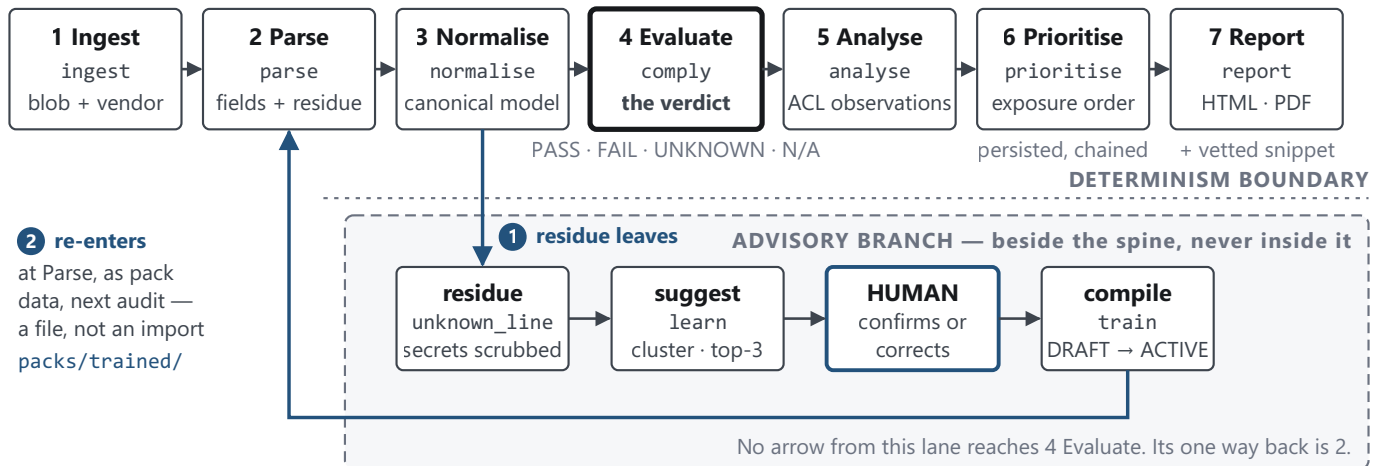
NIRIKSHAK — architecture

AI suggests. Rules decide. Humans confirm.

SIH 2026 · PS 26155 · NTRO · Team Atlantis

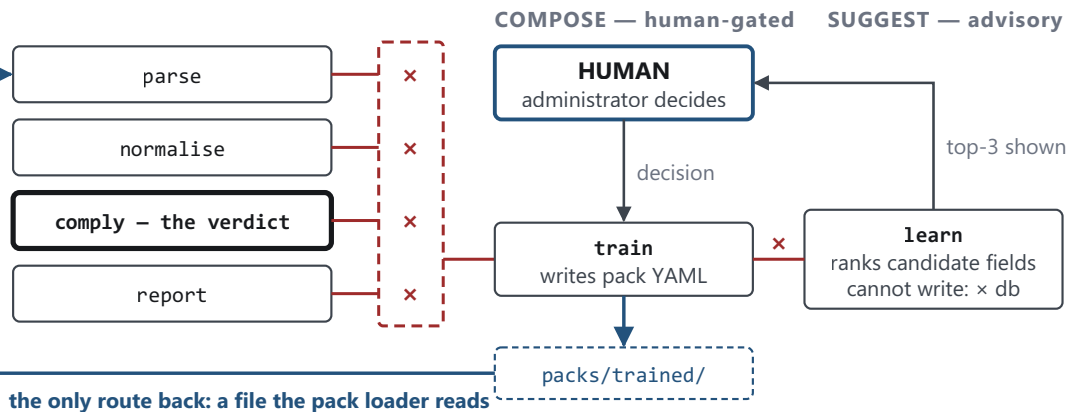
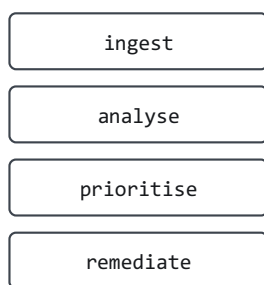
Full reference: docs/architecture.md · decisions: docs/adr/

DETERMINISTIC SPINE — every verdict is produced here, in this order



1 · The pipeline. Seven stages in the order `run_audit_endpoint` (`api/routers/audits.py`) calls them. Unmatched lines leave after normalisation (`csml.residue → record_residue`); a learned mapping returns only as a vendor-pack file, read by `find_pack`.

DECIDE — deterministic



75 forbidden import edges tests/architecture/test_import_rules.py
27 of them touch learn or train: 15 point in, 11 point out, 1 joins them (learn × train).

Suggestion has no value
api/models/training.py

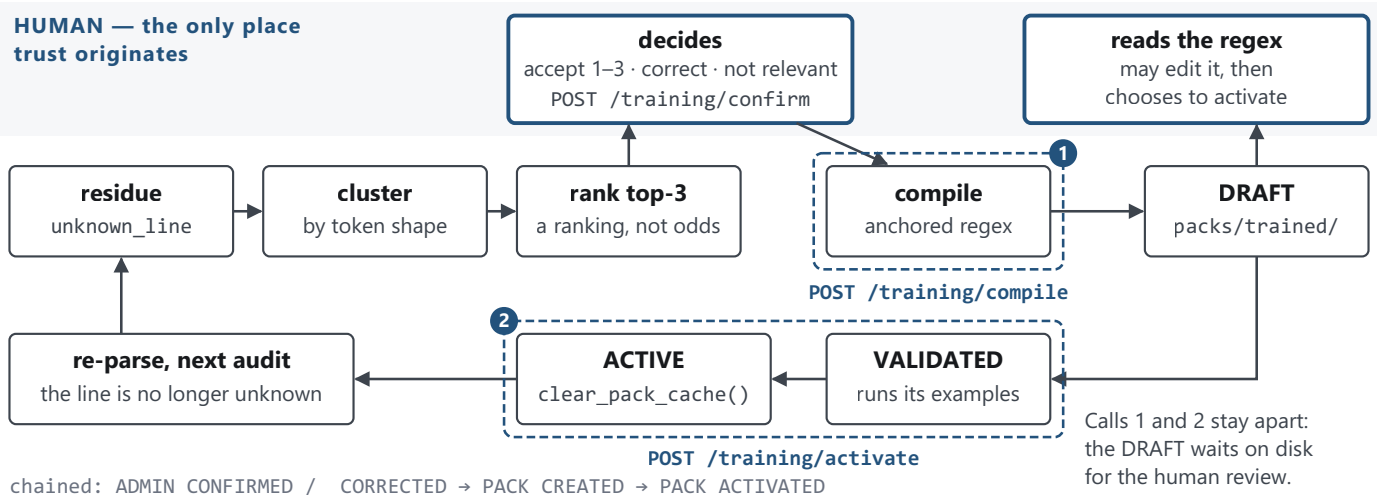
ML imports only in learn
test_learn_boundaries.py

Model may only suggest
audit/0001_initial.sql

Uncalibrated → UNKNOWN
api/models/field.py

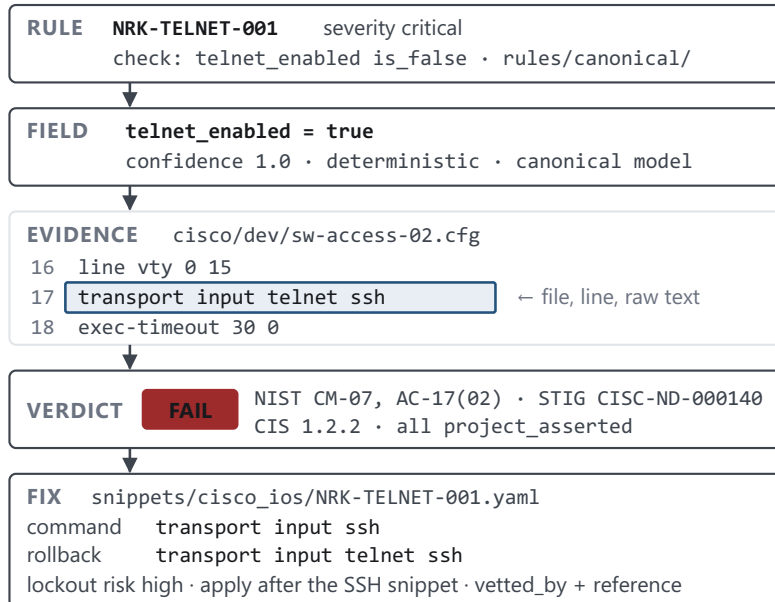
2 · The determinism boundary. Each red stub is an import a test forbids: the verdict package cannot see learn or train, and neither can see it. A suggestion reaches a verdict only as a file that a human caused to exist.

HUMAN — the only place trust originates

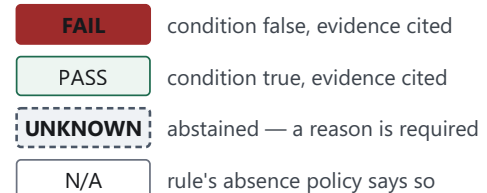


3 · The confirmation loop. confirm → compile_confirmation → activate_draft, all in `api/train/service.py`. The model ranks; an administrator decides, then reads the compiled regex before any configuration is read with it.

ONE FINDING, EXPLODED — a real one, from this commit



FOUR VERDICTS — weight, not colour alone

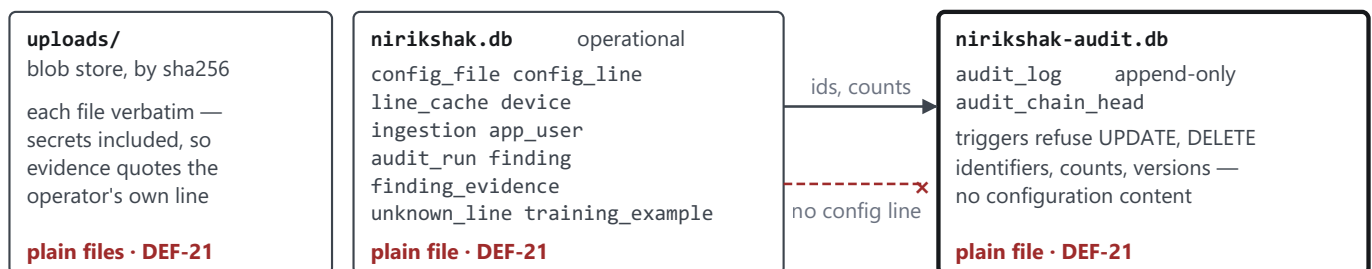


THREE CAUSES OF UNKNOWN — never merged

- 1 · The packs cannot read it**
no_match · unparsed_block
low_confidence
uncalibrated_confidence → training queue
- 2 · Read, but not determinable**
capability_unknown · no_evidence
conflicting_evidence → vendor docs
- 3 · The rule is wrong**
rule_type_mismatch → fix the rule

Same run: NRK-SSH-001 is UNKNOWN, cause 2.

4 · Evidence and verdicts. evaluate_device (api/comply/engine.py) reads only the canonical field; the field carries its line; the fix comes from the vetted library with its rollback and impact. Causes are the eight UnknownReason values (api/models/enums.py), grouped by where the fix lives; cause 1 is exactly Finding.needs_training.



THE CHAIN — each record binds its predecessor



entry_hash = SHA-256(seq, timestamp, actor, action, subject, payload_hash, prev_hash)
 chained: file_ingested · file_rejected · audit_run · admin_confirmed · admin_corrected
 pack_created · pack_activated · pack_rolled_back · report_generated

ai_suggested:
never · DEF-20

5 · Storage and attestation. Two databases so that "the audit store holds no configuration content" is checked by opening one file (ADR 0009, D4). The chain is tamper-evident, not tamper-proof: it is unkeyed, so an attacker with full write access can recompute it (ADR 0008). Hashing: compute_entry_hash, api/models/audit.py.

What NIRIKSHAK does not claim — true at commit 28ab52b

- Encryption at rest.** Required by Rule 6; not built. The databases and uploads/ are plain files (DEF-21).
- A chained record of each AI suggestion.** Required by CLAUDE.md §9; ai_suggested is never appended (DEF-20).
- ISO/IEC 27001.** Not in the selector: a purchased standard, not held. STIG and CIS editions describe Cisco IOS XE 17.x only.
- An official framework mapping.** Every mapping is project_asserted.
- Calibrated confidence or top-3 accuracy.** No line-level ground truth; calibration refuses under 200 samples.
- Absence-aware evaluation on real data.** No pack ships a platform default or capability claim.
- An exposure ranking.** No pack declares management interfaces, so prioritisation abstains.
- Held-out generalisation.** PAN-OS is sealed; XML and JSON modes raise rather than parse.
- Field accuracy.** Every corpus file is hand-written by the team; labels are unreviewed.
- A tamper-proof log.** Tamper-evident only (ADR 0008).
- Serial numbers.** Not present in a configuration export (ADR 0044).
- FAISS or any LLM.** Ranking is an exact cosine scan (api/learn/suggest.py).